Variational Autoencoders for Multi-Omic Transcriptomic and Epigenomic Data

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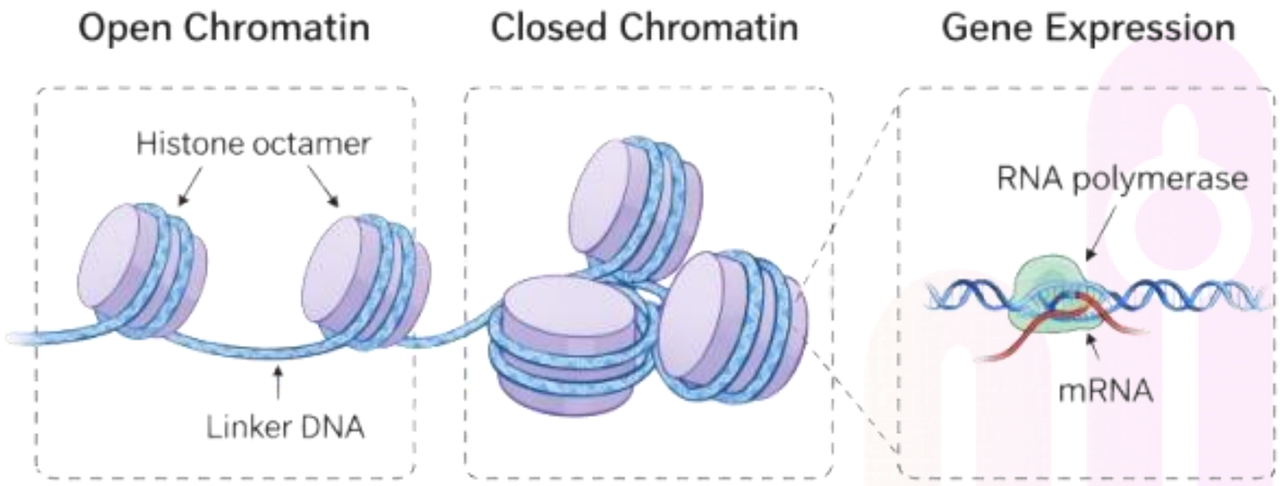
Introduction

Biological Background

- **Gene expression** refers to the production of messenger RNA (mRNA) from DNA, reflecting transcriptional activity.
- **Chromatin accessibility** describes accessibility of DNA regions to regulatory proteins, measured as peaks.
- **Integration of modalities** provides a broader view of cellular regulation.



Introduction

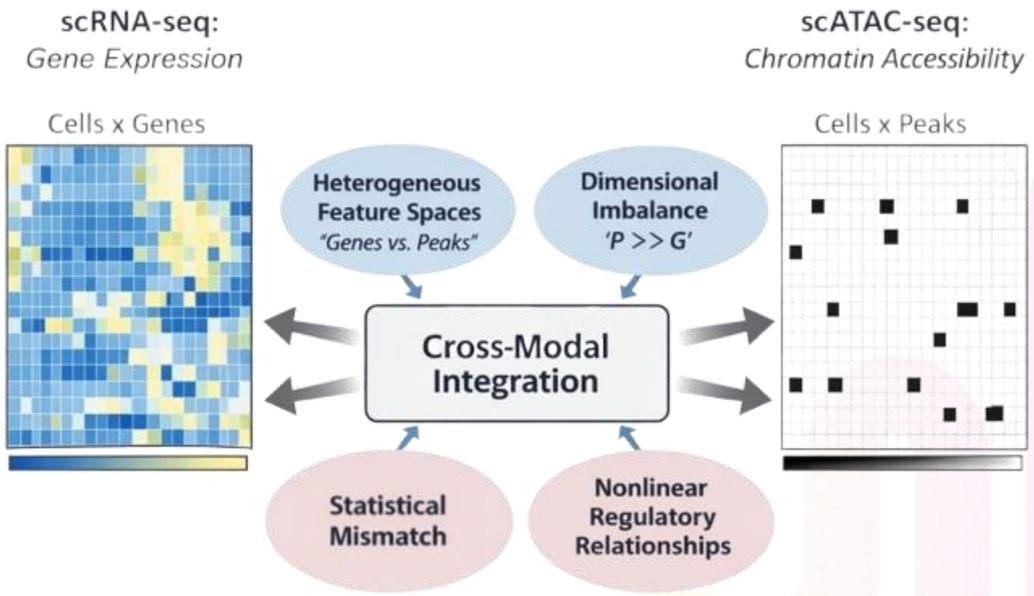
Profiling Technologies

- **Single-cell technologies** measure cellular states in individual cells, rather than at the population level.
- **scRNA-seq** measures gene expression by quantifying mRNA counts for each gene in a cell.
- **scATAC-seq** profiles chromatin accessibility in individual cells by detecting accessible DNA regions, identified as peaks.
- **Paired multi-omic measurements** provide both RNA and ATAC profiles for the same cell.

Introduction

Motivation

- **Problem setting:** RNA and ATAC profiles are represented in different feature spaces, defined by genes and peaks
- **Objective:** To learn gene–peak relationships at single-cell resolution
- **Challenges:** sparsity, high dimensionality, and nonlinear relationships

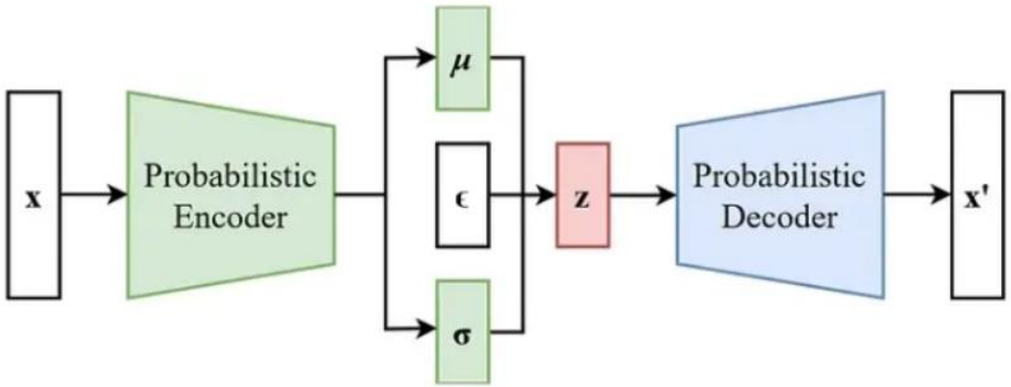


Methods

Modeling Approach

- **Variational Autoencoders (VAEs):** Generative models that map high dimensional data to a probabilistic latent space and reconstruct input

- **VAE-based architectures developed for RNA–ATAC integration:**
  - **GPG:** Sequential Gene–Peak–Gene model
  - **PGP:** Sequential Peak–Gene–Peak model
  - **Dual-VAE:** Modality-specific VAEs with shared latent alignment



Methods

Workflow

- **Datasets:** PBMC 3k and PBMC 10k paired multiome datasets from 10x Genomics, comprising scRNA-seq and scATAC-seq profiles

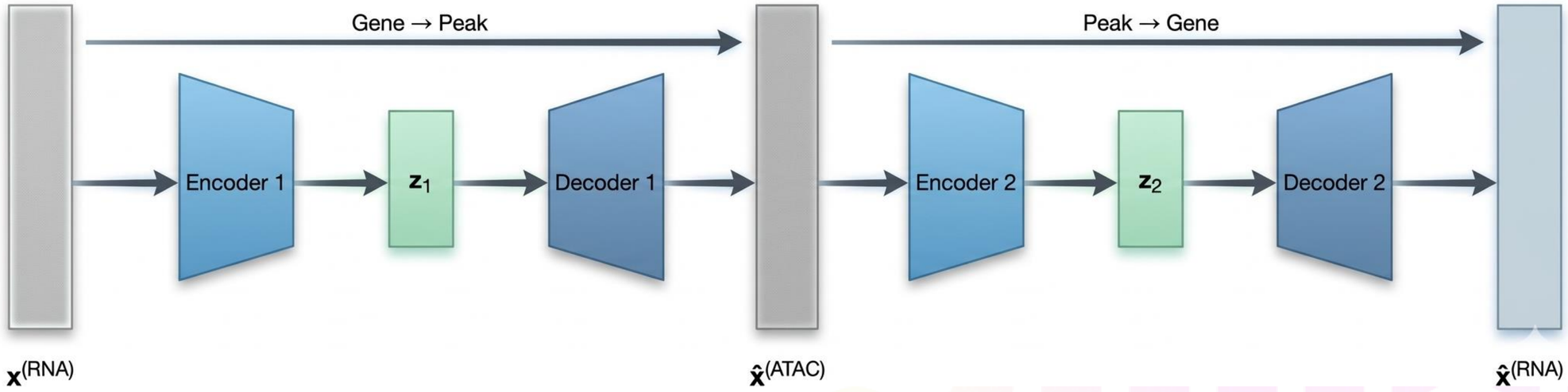
- **Preprocessing:** quality control, filtering, normalization, and paired-cell alignment

- **Modeling objectives:**
  - learning biologically meaningful cross-modal latent representations
  - predicting chromatin accessibility and reconstructing gene expression
  - investigating gene–peak relationships through perturbation-based explainability

Methods

GPG Model

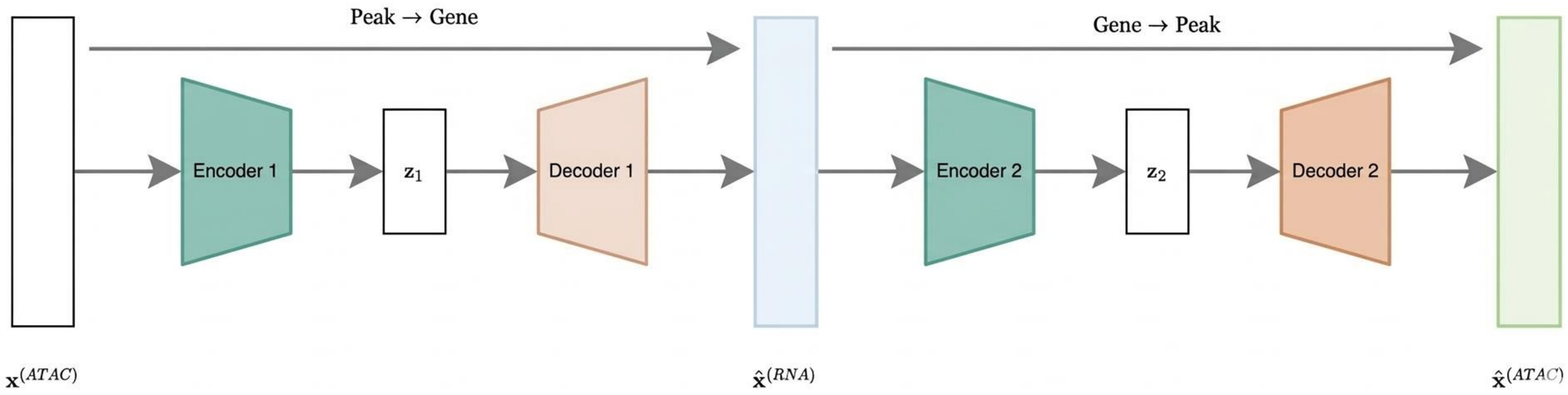
- **Structure:** RNA profile encoded into z_1 and decoded into a predicted ATAC profile, then re-encoded into z_2 and decoded into reconstructed RNA profile
- **Loss Function:** RNA reconstruction + ATAC reconstruction + KL regularization for z_1 and z_2



Methods

PGP Model

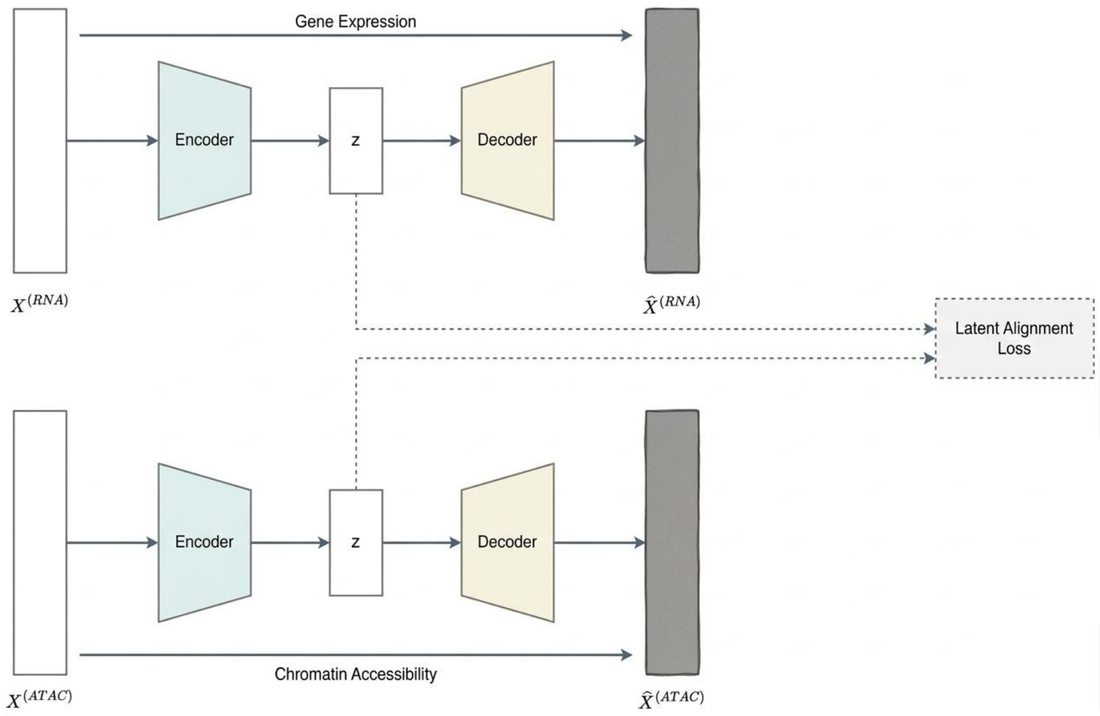
- **Structure:** ATAC profile encoded into z_1 and decoded into a predicted RNA profile, then re-encoded into z_2 and decoded into reconstructed ATAC profile
- **Loss Function:** RNA reconstruction + ATAC reconstruction + KL regularization for z_1 and z_2



Methods

Dual-VAE model

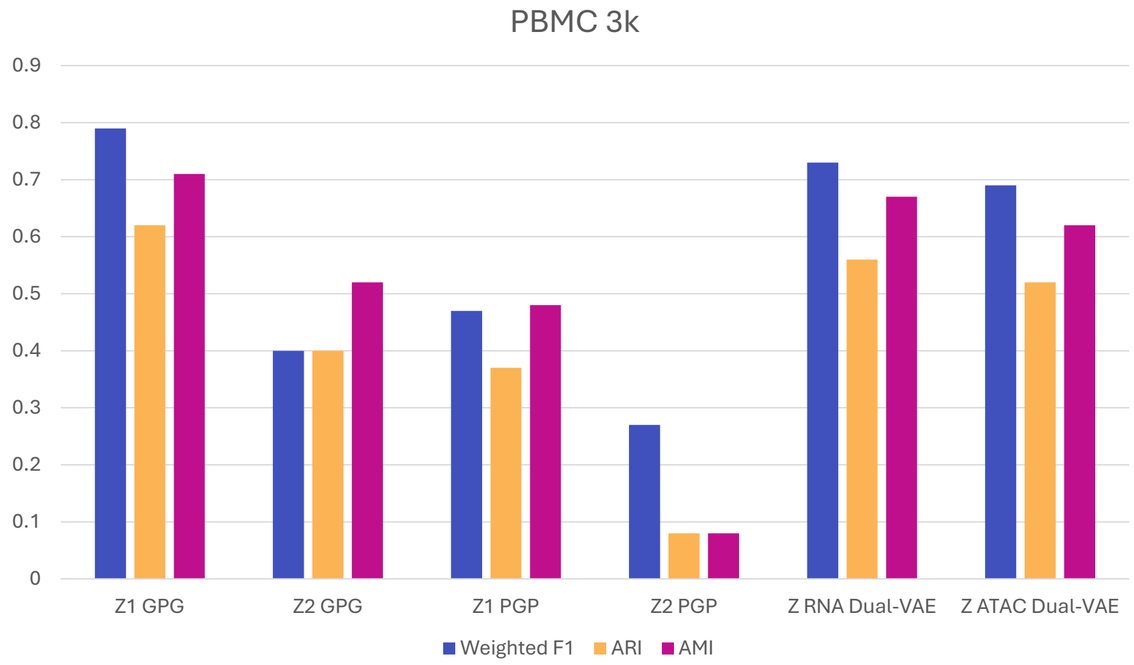
- **Structure:** RNA and ATAC profiles are encoded into separate modality-specific latent spaces and decoded to reconstruct corresponding profiles
- **Shared Alignment:** The two modality-specific latent spaces are coupled through a shared latent alignment term
- **Loss function:** RNA reconstruction + ATAC reconstruction + KL regularization for both branches + latent alignment

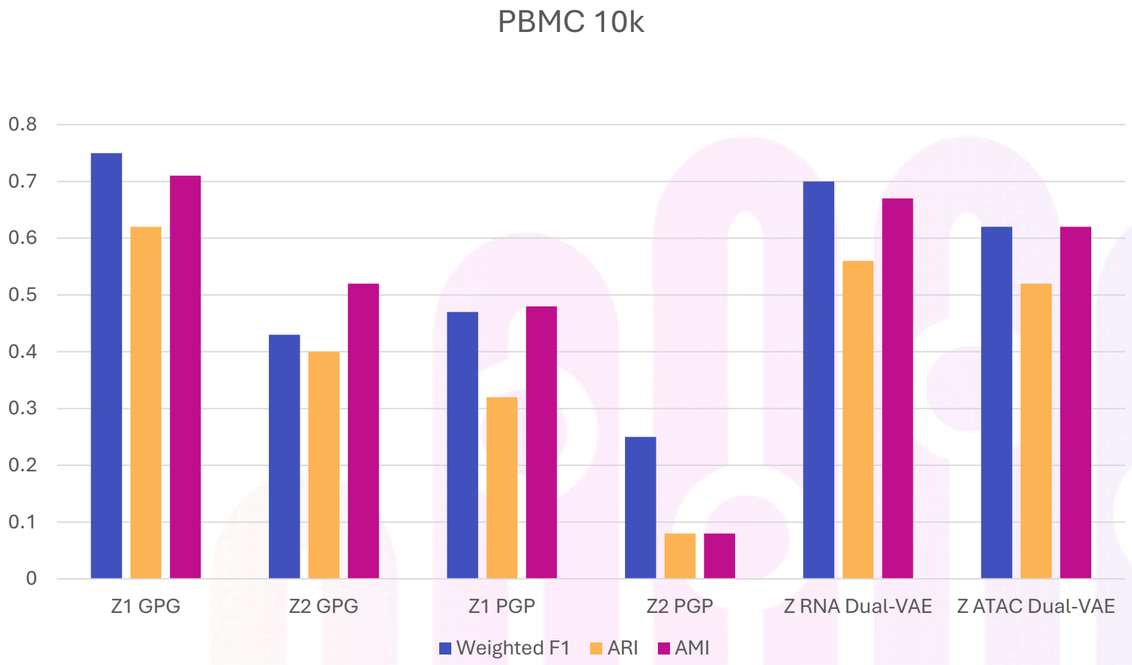


Results

Quantitative Latent Space Evaluation

Prediction of RNA-derived Leiden labels from latent representations

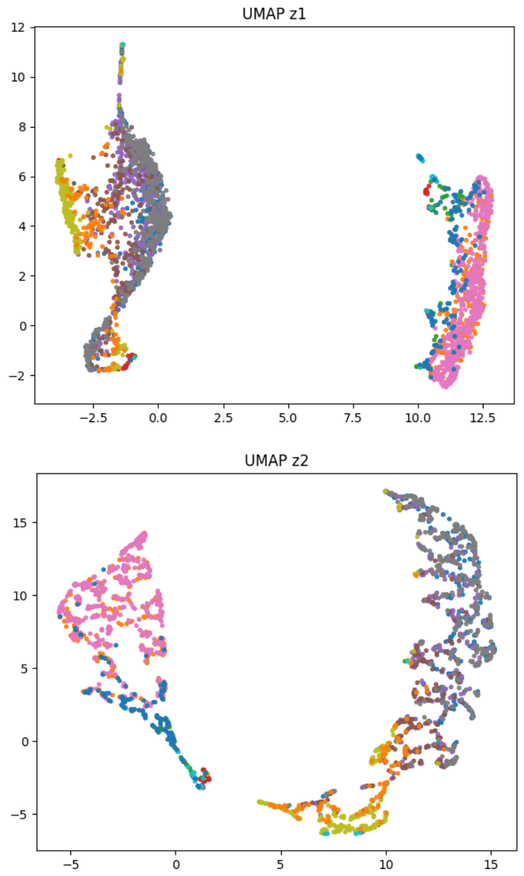




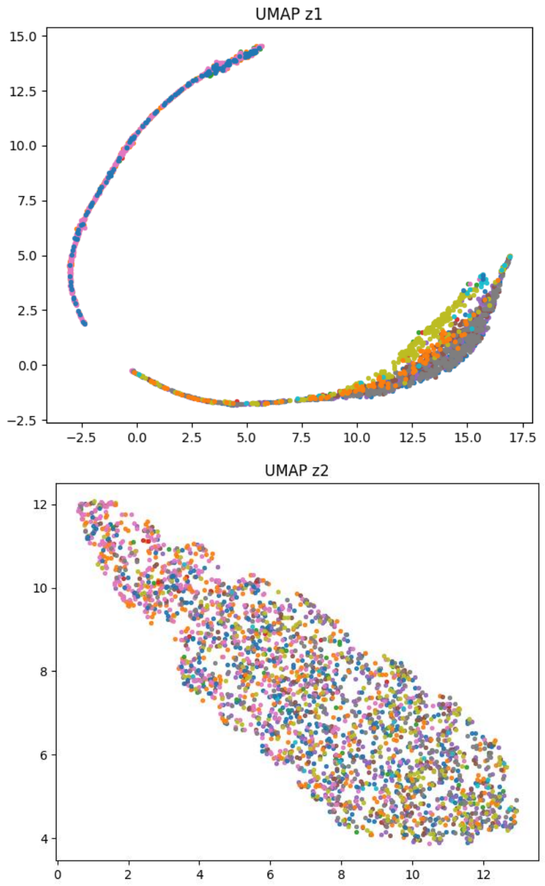
Results

Qualitative Latent Space Evaluation – PBMC 3k

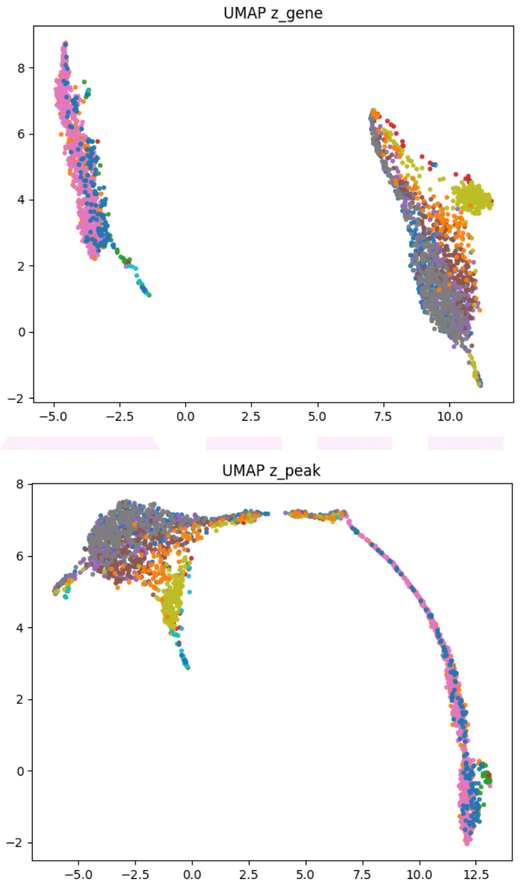
GPG Model:



PGP Model:



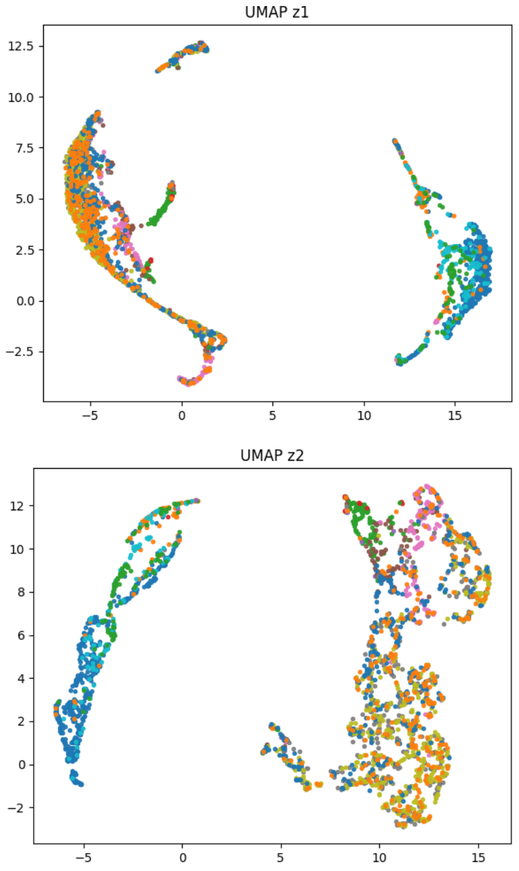
Dual-VAE Model:



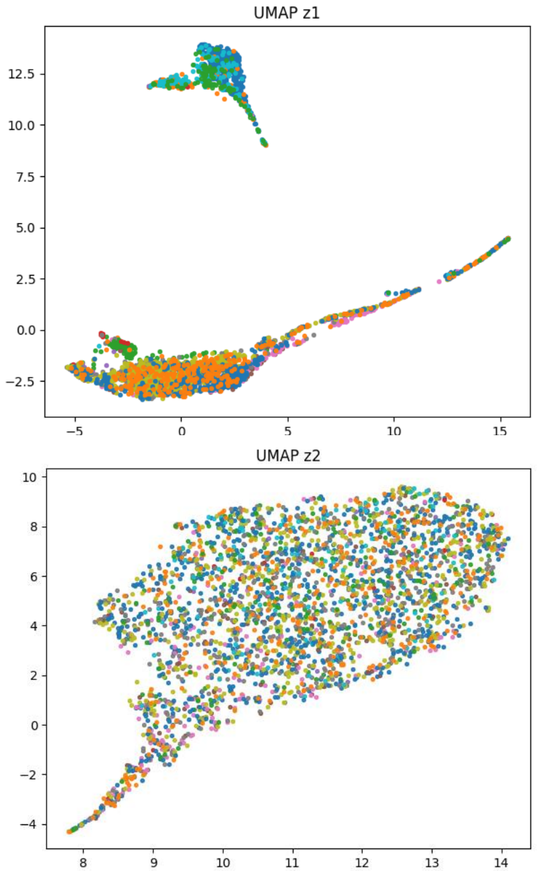
Results

Qualitative Latent Space Evaluation – PBMC 10k

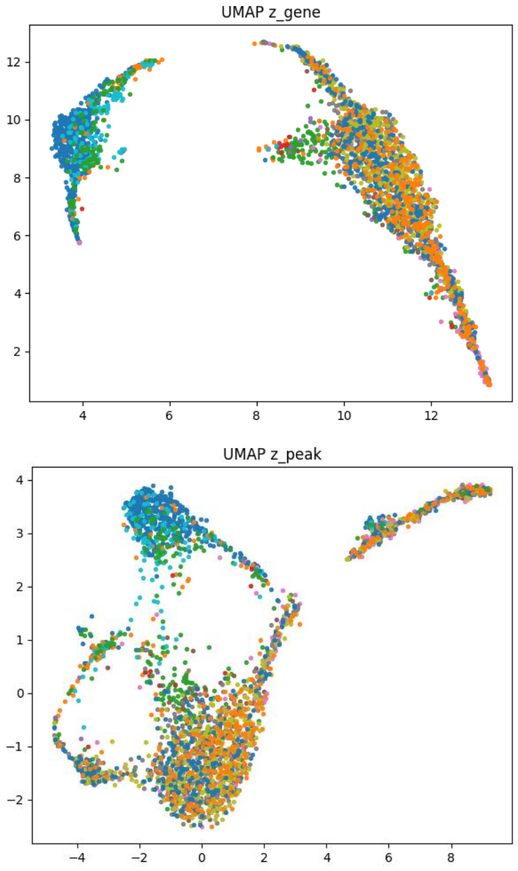
GPG Model:



PGP Model:



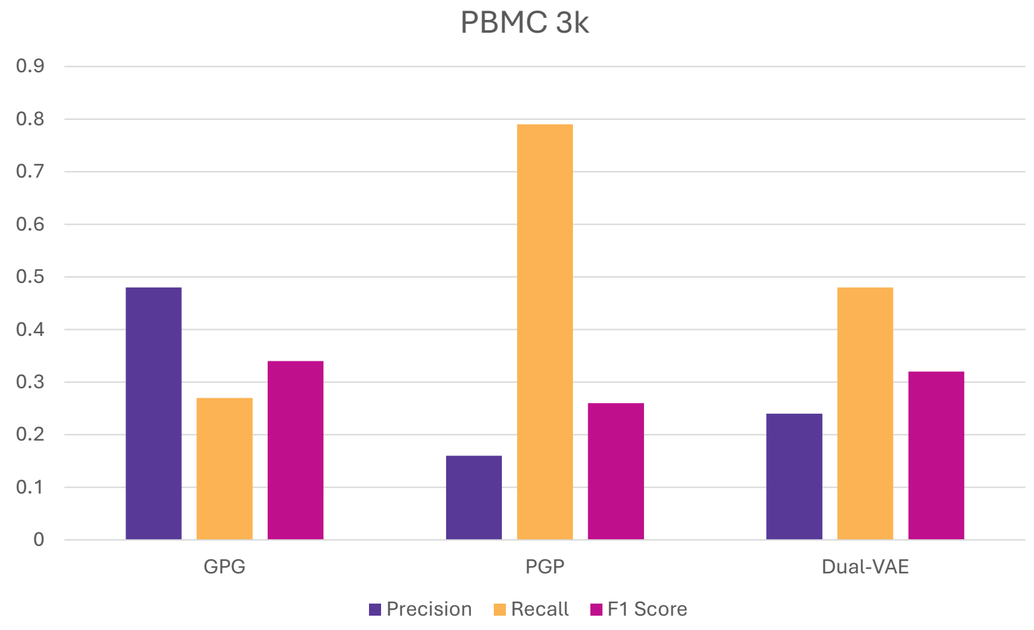
Dual-VAE Model:

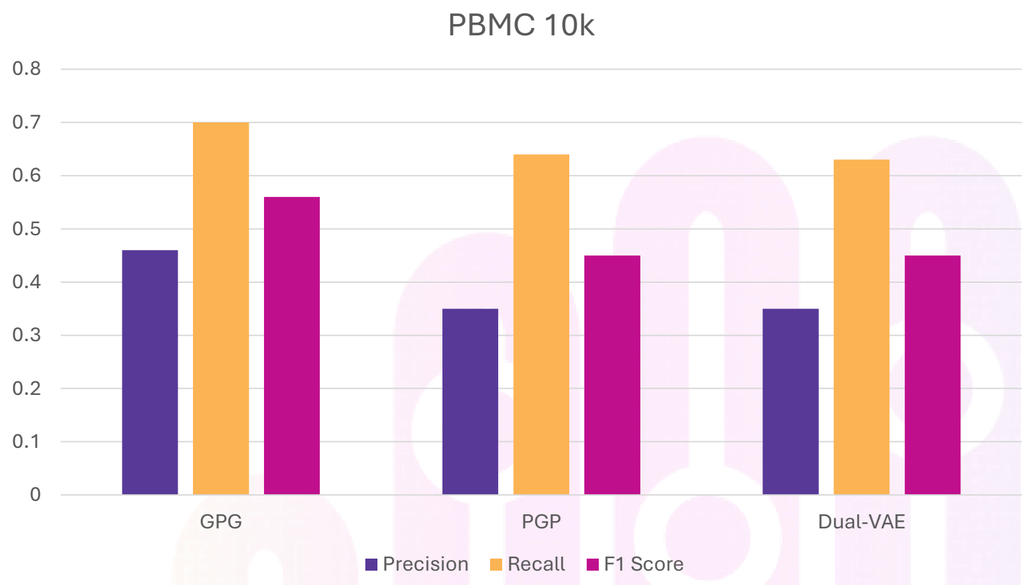


Results

Chromatin Accessibility Prediction

Precision, Recall, and F1-Score for Chromatin Accessibility Prediction across models

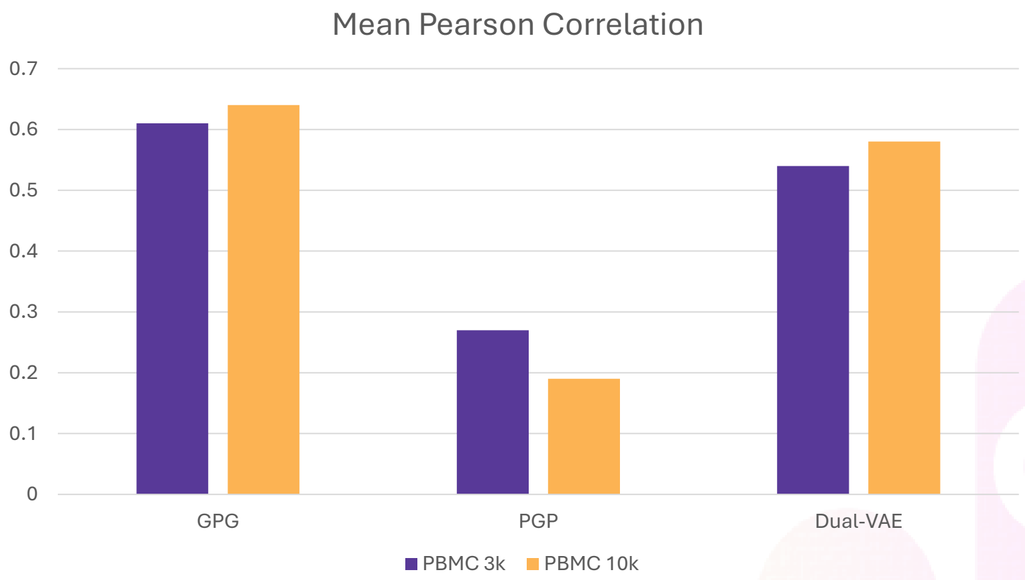




Results

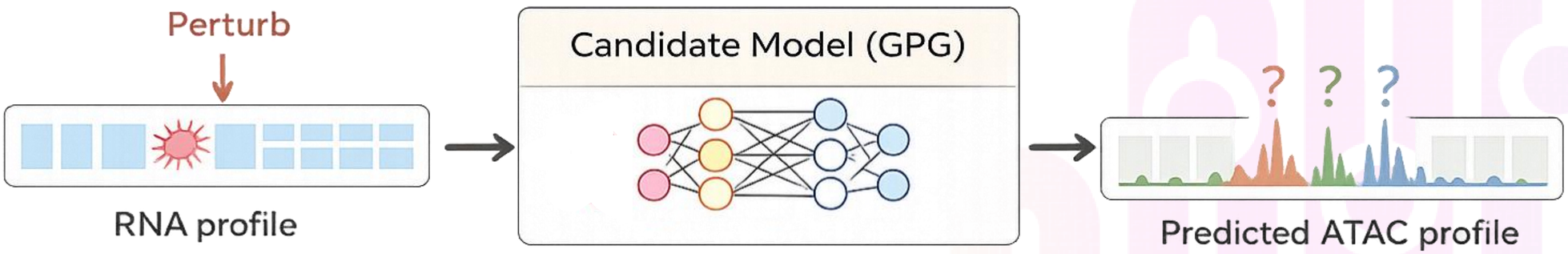
Gene Expression Reconstruction

Mean Pearson Correlation for Gene Reconstruction



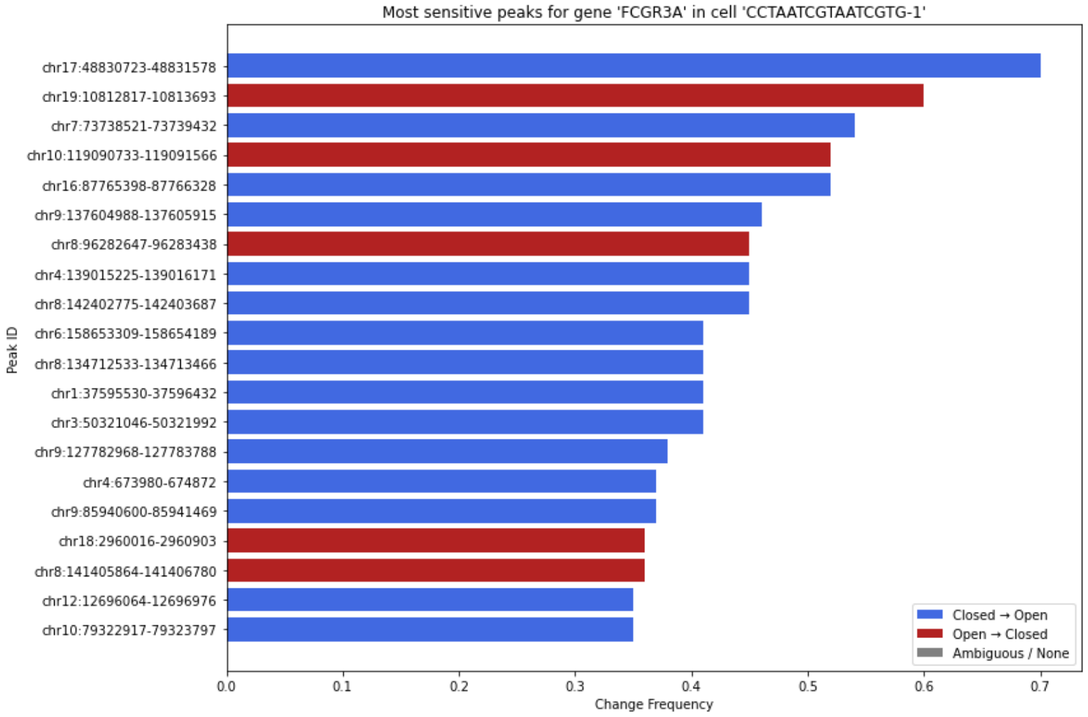
Explainability Analysis

- GPG was selected as the candidate model for explainability analysis on PBMC 10k
- One marker gene was perturbed in a representative cell
- Repeated stochastic forward passes identified peaks changing consistently after gene perturbation

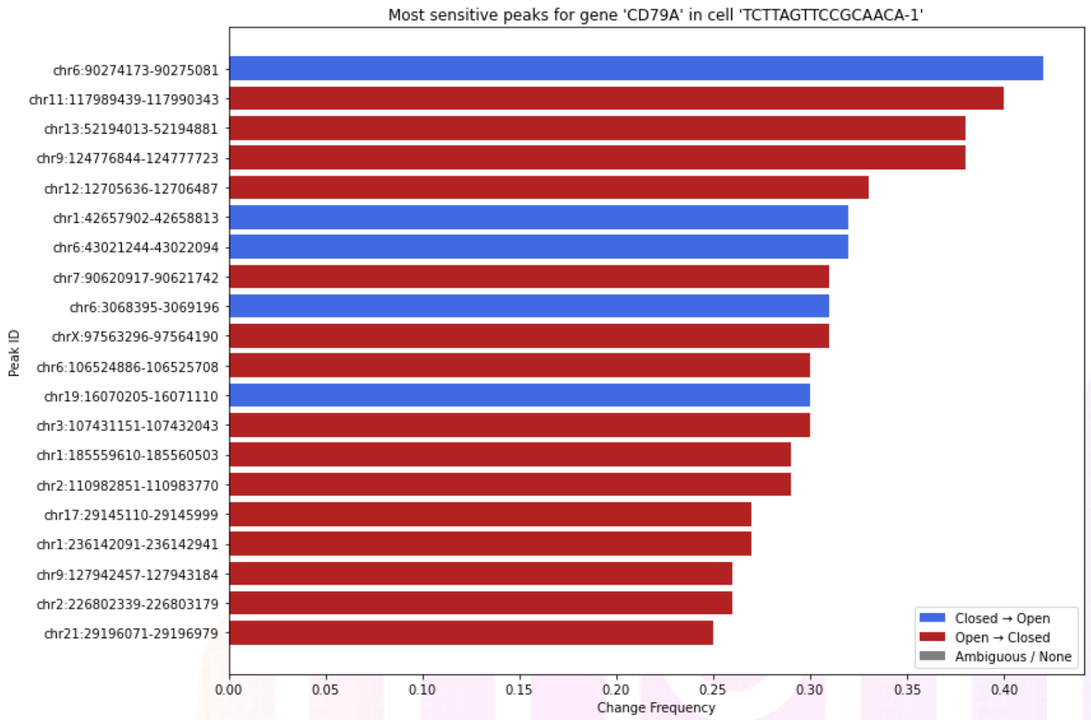


Explainability Analysis

FCGR3A perturbation: 0 → 20



CD79A perturbation: 17 → 0



Conclusion

- **Sequential architectures:** GPG outperformed PGP, indicating that the translation direction used in GPG was more effective than the reverse formulation.
- **Dual-VAE architecture:** Dual-VAE learned informative latent spaces and achieved solid reconstruction results.
- **Interpretability analysis:** The perturbation-based framework supported the analysis of candidate gene–peak relationships in PBMC multiome data.

Future Work

- **Broader evaluation:** Evaluate the models on more diverse paired multiome datasets spanning different cell populations and tissues.
- **Dual-VAE extension:** development toward explicit cross-modal translation and stronger integration between RNA and ATAC modalities.
- **Expanded perturbation analysis:** Extend the explainability framework to multi-gene perturbations, additional cell populations, and comparison with external genomic annotations.

Thank you

for your attention

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